

## 1. Executive Summary

|                         |        | Traffic Summary        |  | Count      |  |
|-------------------------|--------|------------------------|--|------------|--|
| Metric                  | Value  | Total Flows Observed   |  | 234922     |  |
| Total Threats Mitigated | 23     | Total Packets Analyzed |  | 6668606006 |  |
| High Priority           | 13     | Malicious Packets      |  | 6641005434 |  |
| Low Priority            | 10     | True Negatives Passed  |  | 177729     |  |
| Manual Releases         | 0      |                        |  |            |  |
| Manual Blocks           | 0      | Attack Vector          |  | Count      |  |
|                         |        | ICMP Flood             |  | 0          |  |
| ML Metrics              | Value  | SYN Flood              |  | 0          |  |
| False Positives         | 48     | UDP Flood              |  | 16         |  |
| FP Rate                 | 0.01%  | Uncertain              |  | 7          |  |
| IF Precision            | 99.96% |                        |  |            |  |
| IF Recall               | 93.88% | Action                 |  | Count      |  |
| IF F1-Score             | 96.82% | Quarantined            |  | 4          |  |
| IF Accuracy             | 98.41% | Time Ban               |  | 3          |  |
|                         |        | Blackhole              |  | 0          |  |
|                         |        | Released               |  | 7          |  |

## 2. Performance Benchmark

### 2a. Isolation Forest: Anomaly Detection

| Metric                    | Value  | Description                                          |
|---------------------------|--------|------------------------------------------------------|
| Precision                 | 99.96% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 93.88% | Share of actual attacks successfully flagged         |
| F1-Score                  | 96.82% | Balanced measure combining Precision and Recall      |
| Accuracy                  | 98.41% | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.01%  | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 6.12%  | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 93.88% | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 99.99% | Normal traffic correctly identified as safe          |

|                | Predicted: Attack | Predicted: Normal |
|----------------|-------------------|-------------------|
| Actual: Attack | TP 56971          | FN 3717           |
| Actual: Normal | FP 24             | TN 174012         |

### 2b. Random Forest: Attack Classification

| Metric    | Value   | Description                                                  |
|-----------|---------|--------------------------------------------------------------|
| Precision | 100.00% | Share of classified flows assigned the correct attack type   |
| Recall    | 100.00% | Share of attacks of each type that were correctly identified |
| F1-Score  | 100.00% | Balanced measure combining Precision and Recall              |
| Accuracy  | 100.00% | Overall proportion of correct classifications                |

|           | Predicted: SYN | Predicted: ICMP | Predicted: UDP |
|-----------|----------------|-----------------|----------------|
| Act: SYN  | 0              | 0               | 0              |
| Act: ICMP | 0              | 0               | 0              |
| Act: UDP  | 0              | 0               | 56971          |

### 2c. Response Latency

| Metric                   | Value    | Description                                                             |
|--------------------------|----------|-------------------------------------------------------------------------|
| Detection Time           | 251.9 ms | Average time taken to detect an attack                                  |
| Mitigation Response Time | 3.0 ms   | Interval between the anomaly flag and the FlowMod blocking rule install |

2d. Controller Resource Overhead

| Metric                         | Value | Description                                                           |
|--------------------------------|-------|-----------------------------------------------------------------------|
| Controller CPU (Baseline)      | 0.99% | Ryu controller CPU usage under normal traffic                         |
| Controller CPU (Active Attack) | 1.99% | Ryu controller average CPU usage during a DDoS simulation             |
| Controller CPU (Mitigation)    | 3.19% | Ryu controller CPU usage during simultaneous detection and mitigation |

3. Offences Summary

| Source IP | Sessions | Max Offences | First Seen          | Last Seen           |
|-----------|----------|--------------|---------------------|---------------------|
| 10.0.0.22 | 3        | 3            | 2026-09-15 20:11:48 | 2026-09-15 20:13:10 |
| 10.0.0.20 | 2        | 2            | 2026-09-15 20:11:45 | 2026-09-15 20:12:55 |
| 10.0.0.21 | 2        | 2            | 2026-09-15 20:11:46 | 2026-09-15 20:12:56 |

#### 4. Chronological Mitigation Log

| Timestamp           | Source IP | Class | Vector    | Confidence | Priority | Action        |
|---------------------|-----------|-------|-----------|------------|----------|---------------|
| 2026-09-15 10:34:56 | 10.0.0.20 | DDoS  | Uncertain | -          | Low      | Quarantined   |
| 2026-09-15 10:34:56 | 10.0.0.21 | DDoS  | Uncertain | -          | Low      | Quarantined   |
| 2026-09-15 10:34:57 | 10.0.0.22 | DDoS  | Uncertain | -          | Low      | Quarantined   |
| 2026-09-15 10:35:06 | 10.0.0.20 | DDoS  | UDP Flood | 75.9%      | High     | Time Ban (1m) |
| 2026-09-15 10:35:06 | 10.0.0.21 | DDoS  | UDP Flood | 77.9%      | High     | Time Ban (1m) |
| 2026-09-15 10:35:07 | 10.0.0.20 | DDoS  | UDP Flood | 83.3%      | High     | Time Ban      |
| 2026-09-15 10:35:07 | 10.0.0.21 | DDoS  | UDP Flood | 77.9%      | High     | Time Ban      |
| 2026-09-15 10:35:07 | 10.0.0.22 | DDoS  | UDP Flood | 86.4%      | High     | Time Ban (1m) |
| 2026-09-15 10:35:08 | 10.0.0.22 | DDoS  | UDP Flood | 86.4%      | Critical | Time Ban      |
| 2026-09-15 10:36:07 | 10.0.0.20 | DDoS  | UDP Flood | 86.6%      | High     | Released      |
| 2026-09-15 10:36:07 | 10.0.0.21 | DDoS  | UDP Flood | 79.1%      | High     | Released      |
| 2026-09-15 10:36:08 | 10.0.0.22 | DDoS  | UDP Flood | 86.4%      | High     | Released      |
| 2026-09-15 10:36:18 | 10.0.0.20 | DDoS  | UDP Flood | 86.6%      | Critical | Time Ban (2m) |
| 2026-09-15 10:36:19 | 10.0.0.22 | DDoS  | UDP Flood | 86.4%      | Critical | Time Ban (2m) |
| 2026-09-15 10:41:51 | 10.0.0.6  | DDoS  | Uncertain | 48.2%      | Low      | Released      |
| 2026-09-15 10:41:59 | 10.0.0.21 | DDoS  | UDP Flood | 83.2%      | High     | Time Ban (2m) |
| 2026-09-15 10:51:59 | 10.0.0.2  | DDoS  | Uncertain | 61.1%      | Low      | Released      |
| 2026-09-15 11:10:35 | 10.0.0.10 | DDoS  | Uncertain | 48.4%      | Low      | Released      |
| 2026-09-15 11:10:36 | 10.0.0.8  | DDoS  | Uncertain | 64.1%      | Low      | Released      |
| 2026-09-15 11:10:45 | 10.0.0.9  | DDoS  | UDP Flood | 92.4%      | High     | Time Ban (1m) |
| 2026-09-15 11:12:07 | 10.0.0.8  | DDoS  | UDP Flood | 90.9%      | High     | Time Ban (1m) |
| 2026-09-15 11:22:45 | 10.0.0.6  | DDoS  | UDP Flood | 85.1%      | High     | Time Ban (1m) |
| 2026-09-15 11:24:20 | 10.0.0.6  | DDoS  | UDP Flood | 92.4%      | High     | Quarantined   |

#### 5. IP Attack History (Completed Sessions)

| Source IP | Vector    | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|-----------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.22 | UDP Flood | 0.6368   | 78.8% | Critical | Phase 2 | 3        | 11s      | 2026-09-15 20:13:10 | Attack persisted - escalating to time ban |
| 10.0.0.22 | UDP Flood | 0.6007   | 77.7% | High     | Phase 2 | 1        | 1m 11s   | 2026-09-15 20:12:59 | Ban Expired                               |
| 10.0.0.21 | UDP Flood | 0.6096   | 79.8% | High     | Phase 2 | 2        | 1m 10s   | 2026-09-15 20:12:56 | Attack persisted - escalating to time ban |
| 10.0.0.20 | UDP Flood | 0.6285   | 77.5% | High     | Phase 2 | 2        | 1m 10s   | 2026-09-15 20:12:55 | Attack persisted - escalating to time ban |
| 10.0.0.22 | UDP Flood | 0.6275   | 77.7% | High     | Phase 2 | 1        | 10s      | 2026-09-15 20:11:58 | Attack persisted - escalating to time ban |

| Source IP | Vector    | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|-----------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.21 | UDP Flood | 0.6186   | 79.8% | High     | Phase 2 | 1        | 10s      | 2026-09-15 20:11:56 | Attack persisted - escalating to time ban |
| 10.0.0.20 | UDP Flood | 0.6346   | 75.6% | High     | Phase 2 | 1        | 10s      | 2026-09-15 20:11:55 | Attack persisted - escalating to time ban |

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE